

High-Level Architecture Overview

Referencing your diagram, this PoC integrates:

- **Azure Data Factory (ADF)** for orchestration.
- **Blob Storage** as a central staging layer.
- **Azure Function Apps** for custom processing and delivery logic.
- **On-prem delivery via SFTP, Windows Share, NFS, MSSQL.**
- **Power BI and Power Query** for reporting and analytics.
- **A Reporting Management UI** for self-service configuration.



Design Overview



1. Feed Configuration and Self-Service Management

- **Frontend:** PowerApps as a Reporting Management UI.
- **Backend:**
 - Stores metadata in Azure SQL DB (or Cosmos DB for scale).
 - Feed configuration includes:
 - Source details (semantic model or pre-defined datasets).
 - Output format (CSV, Excel, JSON).
 - Delivery method (SFTP, Email, Share, NFS, MSSQL).
 - Schedule (CRON-based, ad hoc).
 - Recipients and metadata validations.



2. Data Orchestration (ADF)

- **Trigger:** ADF pipeline triggers based on:
 - Scheduled intervals (via ADF triggers or Autosys).
 - Manual trigger from Reporting Management UI.
- **Ingest & Process:**
 - Pulls data via Function App or directly from source/semantic model.
 - Applies transformations using:
 - ADF Data Flows (for simple logic).
 - Function Apps (for complex templating or conditional logic).

- **Stores output in Blob Storage** in designated format.



3. Delivery Logic (Function Apps)

- ADF triggers **Azure Function** with metadata:
 - Destination type.
 - Output file location (Blob URI).
 - Recipients or path info.

Supported Deliveries:

Channel	Method
Email	SendGrid or Graph API w/ templating
SFTP	Azure Function uses SFTP library or MFT connector
Windows Share / NFS	Function uses hybrid worker or private link
Database	Direct insert via on-prem IR or Function
API	HTTP Post via Function App



4. Monitoring & Auditing

- Each feed execution logs into **Azure Log Analytics / SQL**:
 - Trigger time, source, output path, delivery status.
 - Errors (transformation/delivery).
- Exposed via:
 - Power BI dashboard for Ops visibility.
 - Email alerts on failure (optional autosys trigger or Function App).



Implementation Plan (PoC)

Phase	Tasks	Tools/Tech
1	Setup ADF pipeline: Create pipeline w/ dummy dataset and transformation	ADF
2	Blob Storage: Setup container for intermediate and final files	Blob
3	Create Azure Function: To deliver files via Email, SFTP, and Windows Share	Function App
4	Develop PowerApps UI: Feed configuration form, validate, and store config in SQL	PowerApps, Azure SQL

Phase	Tasks	Tools/Tech
5	Metadata-Driven Pipelines: Fetch feed config from SQL and dynamically execute flows	ADF
6	Monitoring Framework: Setup logging tables and Power BI dashboard	Power BI, Azure Monitor
7	MVP Test Feed: End-to-end run: ingest → transform → store → deliver → log	All



PoC Test Case

Scenario: Deliver regulatory report (CSV format) via SFTP every weekday at 6 AM.

- User configures feed** in Reporting UI:
 - Source: Dataset A
 - Format: CSV
 - Destination: SFTP
 - Schedule: M-F, 6:00 AM
 - Email notification on failure
 - ADF triggers pipeline** on schedule.
 - Function App:**
 - Reads config, transforms data, formats to CSV.
 - Delivers to SFTP.
 - Logs entry** to SQL and flags success/failure.
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Extensibility Notes

- Adding new delivery channels: Implement plug-in logic in Azure Functions.
- Adding new formats: Extend transformation templates or logic in ADF/Function.
- Integrate holiday calendars using Azure Calendars or JSON configs in SQL.
- Secure access via Managed Identities, Private Endpoints, Key Vault.